

UNIT 8

THE FUTURE OF COMPUTING



Connect to the topic

1. Look at the photo.
What do you think the person is doing?
2. How is this screen different from the ones we use today?
3. Do you think computers will become fully touch-based in the future? Why or why not?
4. How do you imagine computers will look in 50 years?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video that predicts how technology will change from 2025 to 2049.

1. What technology from the video did you find the most interesting? Why?
2. The video shows AI helping people control their dreams. Would you like to try this? Why or why not?
3. What are the advantages and dangers of having robots as police officers?
4. Imagine you are living in 2049. What job do you have?
5. Do you think future technologies will make people happier? Why or why not?
6. If you could visit the future for one day, what would you like to see?



UNIT 8.1 HOLOGRAPHIC DISPLAYS

READING AND VOCABULARY

Task 1. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

float (v) – to appear to stay in the air without visible support.

flat (adj) – having a level, two-dimensional surface.

interference (n) – the combination of two or more waves resulting in a new wave pattern.

light beam (phr) – a narrow stream of light energy.

emitter (n) – a device that sends out light, sound, or other signals.

waveguide (n) – a structure that directs the flow of light or other waves.

rotate (v) – to turn around an axis or center.

depth-sensing (adj) – detecting how far objects are from a sensor, often used in 3D cameras.

gesture recognition (phr) – interpreting human movements (e.g. hands) as input commands.

rendering engine (phr) – software that generates images or 3D models in real time.

1. Traditional screens are _____, showing images that lack depth or dimension.
2. With _____ software, users can control the display just by moving their hands.
3. A hologram can _____ in the air, creating the illusion of a physical 3D object.
4. A laser _____ is used to direct precise streams of light into a holographic projection.
5. A/an _____ system helps the computer understand where your hand is in 3D space.
6. The hologram will _____ as you walk around it, showing different angles.
7. The image is created through the _____ of laser light beams.
8. In holographic displays, a/an _____ produces the light needed to generate the 3D image.
9. The system's _____ processes visual data to produce interactive 3D models.
10. A/an _____ directs the laser light to specific angles to build the hologram.

Task 2. READ FOR DETAILS. Read the text and answer the questions below.

Inside Holographic Display Systems

Imagine viewing a 3D image floating in mid-air – no glasses, no screen, just light forming a dynamic image in front of your very eyes. It's not science fiction. It's a holographic display technology, and it may change how we work, learn, and communicate in the coming years. Unlike traditional screens displaying two-dimensional, flat images, holographic screens create actual 3D images that appear to float in the air. The images can be viewed at different angles without putting on special glasses or headsets.

At the heart of this technology is the hologram – a three-dimensional projection produced by interference of light beams. Advanced systems utilize laser emitters, spatial light modulators, and optical waveguides to control the shape and character of the light. The final product is an incredibly detailed picture that may be caused to move, rotate, and even be controlled by user commands.

Holographic screens are part of the broader category of mixed reality, where virtual content is combined with real-world content. Some systems incorporate depth-sensing cameras and gesture recognition software, which allow users to use their hands to interact with holograms. Such systems process input data in graphics processing units and rendering engines in real-time.

The technology itself is also changing. Today, holographic screens are expensive and used for research, product presentations, or entertainment. But as the technology matures, they may make their way into offices, classrooms, and smartphones too. In commerce and industry, the uses are gigantic. Engineers can view 3D CAD models in mid-air, medical professionals can visualize anatomical structures during surgery, and architects can present interactive building designs to

customers. In computer training, holographic screens are able to make code visualizations and system architecture real, enabling students to comprehend complicated arrangements through 3D interaction.

However, the technology also comes with its setbacks. Current holographic systems require strong processing power, bandwidth, and storage capacity. Hardware is still expensive, and resolution is pretty low compared to high-end 2D screens. But optical computing, AI-based rendering, and light field technology all develop exponentially, which will ultimately yield higher performance and reduced costs. Microsoft (with HoloLens), Looking Glass, and Leia Inc. are leading the way. As progress continues, holographic displays may be a part of everyday IT environments that span from smartphones and AR headsets to meeting devices and team collaboration spaces.

(adapted from <https://www.telefonica.com/en/communication-room/blog/hologram-created/>)

1. What is the main advantage of holographic displays compared to traditional screens?
2. What forms the core of holographic display technology?
3. Why are special glasses or headsets not needed for viewing holograms?
4. What components help holograms interact with users?
5. What are current uses of holographic displays?
6. How might doctors use holographic displays in the future?
7. What are some challenges facing this technology?
8. Name two companies mentioned that are developing holographic display systems.

Task 3. COLLABORATE. Work in pairs. Study a list of components commonly found in holographic display systems. Sort the terms into the correct categories based on their function.

Components of a holographic system

Category	Description	Components
Light source	generates and controls the light that creates the hologram	
Image processor	handles data and transforms it into a visual 3D image	
Projection/display system	sends light in a specific way to create the floating image	
Interaction interface	lets the user interact with the hologram	
User output	what the user actually sees or experiences	

laser emitter	optical waveguide	floating 3D image
graphics processing unit (GPU)	motion sensor	rendering engine
light field generator	real-time rendering software	holographic projection lens
gesture recognition system	spatial light modulator	depth-sensing camera

Task 4. WORK WITH WORDS. Fill in the gaps with the words from the box.

Volumetric Holographic Displays

throughput ▪ angles ▪ emission ▪ depth ▪ occupy ▪ waveguides ▪
rely ▪ entirely ▪ rotating ▪ spatial

Volumetric holographic displays are a new type of display technology that creates 3D images you can see from all sides without special glasses. Unlike flat screens or regular holograms, these

systems produce images that 1) _____ real volume in space. You can walk around them and view them from different 2) _____, just like a physical object.

To do this, the system uses a laser light source, often combined with 3) _____ mirrors, diffusers, or even photo-reactive materials. A key component is the volume pixel, or voxel, which holds 4) _____, height, and width information. These voxels are created in layers using optical 5) _____, light field technology, or swept-surface projection methods.

Precise synchronization between motion and light 6) _____ is essential for the image to appear stable and coherent. Many systems include 7) _____ light modulators to control how and where the light forms the image. Because of the high data 8) _____ and computing needs, volumetric displays often 9) _____ on advanced graphics processing units and real-time software to maintain performance.

Though still in development, volumetric holography represents a major step forward in display technology as they offer a physically present 3D image formed 10) _____ by light.

WATCHING

Task 5. EXPLORE THE WORDS. Match the words from the video to their definitions.

1. field of view	a) a strong, focused light used to shine on something specific
2. in mid-air	b) the area you can see at one time without moving your eyes or head
3. sub-pixels	c) able to be seen
4. reflection	d) to change direction or shape
5. transmission	e) tiny parts of a screen pixel that work together to show color
6. spotlight	f) complete or whole
7. entire	g) floating or appearing to be in the air without touching anything
8. viewable	h) the use of technology to make a person seem present in a different place
9. bend	i) when light bounces off a surface, like a mirror
10. telepresence	j) the process of light passing through something

Task 6. WATCH FOR DETAILS. Scan the QR code and watch the video “Hologram Technology (part 1)”. Choose the correct answers to the questions.



- What technology do smartphones use to create images?
 - Holographic projection
 - White light lasers
 - Sub-pixels on a screen
 - Floating light beams
- How is a reflection hologram different from a transmission hologram?
 - It can be viewed in daylight or under a spotlight
 - It only works at night
 - It requires laser light to be seen
 - It has no depth and appears flat
- What creates the 3D effect in a transmission hologram?
 - Projecting images from two cameras
 - Sub-pixel manipulation
 - Light bending as it passes through the hologram
 - A spinning LED matrix
- Who is considered the inventor of holography?
 - Nikola Tesla
 - Yuri Danishek
 - Stephen A. Benton
 - Denis Gabor

5. Why wasn't the first hologram created until 1962?
 - A. Computers weren't fast enough
 - B. Laser technology didn't exist
 - C. There were no displays available
 - D. The internet wasn't invented
6. What was Dr. Yuri N. Danishek's contribution to holography?
 - A. He invented the first 2D screen
 - B. He made holograms visible in daylight using ordinary bulbs
 - C. He developed smart glasses for viewing holograms in daylight
 - D. He discovered how to transmit holograms through Wi-Fi
7. What was unique about Stephen A. Benton's white light transmission holography?
 - A. It allowed cheap mass production of holograms
 - B. It created motion in holograms
 - C. It allowed for 3D audio
 - D. It worked only with specialized hologram projectors
8. What does ARHT Media aim to create?
 - A. Holographic shopping assistants
 - B. Holographic animals for circuses
 - C. Virtual reality gaming systems
 - D. Real-time holographic telepresence
9. Why are holographic animals being developed for circuses?
 - A. To reduce costs
 - B. To attract younger audiences
 - C. To replace real animals and improve safety
 - D. To create digital zoos
10. What is Pulse Evolution known for?
 - A. Creating holographic mobile phones
 - B. Developing holographic weapons
 - C. Designing 3D printed body parts for medical use
 - D. Creating live holographic performances of artists

Task 7. WATCH FOR MAIN IDEA. Scan the QR code and watch the video "Hologram Technology (part 2)". Put the five application areas in the order they were mentioned in the video.

- Design and creativity
- Real-life gaming
- Medical use
- Military use
- Education



Task 8. WATCH FOR DETAILS. Watch the video again. Complete the sentences with one word or short phrase from the video.

1. Holograms give soldiers a chance to preview the _____.
2. Designers usually create 3D objects on a _____ screen.
3. Medical students can practice _____ on a hologram.
4. Holograms in classrooms make learning more _____.
5. A holographic game might involve fighting a _____ in your living room.
6. Holograms could help visualize _____ in classrooms.
7. Real-life gaming would allow you to build in _____ with your friend.
8. Holographic design would feel like creating objects out of thin _____.

Task 9. WATCH FOR DETAILS. Decide whether the statements below are True, False, or Not Given according to the video.

1. Holograms may replace paper textbooks in schools.	True/ False/ Not Given
2. All military holograms need internet access to work.	True/ False/ Not Given
3. Microsoft is developing real-life holographic gaming.	True/ False/ Not Given
4. Holograms could be used in Minecraft.	True/ False/ Not Given
5. Designers already use 3D holograms daily.	True/ False/ Not Given
6. Students would be less bored in hologram-based classes.	True/ False/ Not Given
7. Holograms may help new doctors practice without harming real patients.	True/ False/ Not Given
8. Holograms are already used in military missions.	True/ False/ Not Given

Task 10. COLLABORATE. Work in pairs or groups. Rank the 5 holographic use cases (military, design, medicine, education, gaming) from most to least useful. Give reasons. Then answer the questions.

- Which will help more people?
- Which is more fun?
- Which is easier to develop with today's technology?

LISTENING

Task 11. LISTEN FOR DETAILS. Scan the QR code and listen to the podcast "Holograms on the Highway".

- What makes Hyundai Mobis' new display different from traditional HUDs?
 - It uses regular LED panels on the dashboard.
 - It covers only a small part of the windshield.
 - It replaces the steering wheel with a touchscreen.
 - It uses the full windshield and creates 3D images.
- What does "HUD" stand for?
 - Hologram Utility Display
 - Heads-Up Display
 - Hologram User Dashboard
 - High-Usage Device
- What technology allows the holograms to appear on the windshield?
 - HOE film and projection system
 - Touch sensors and mobile apps
 - LCD monitors in the car's roof
 - Mirrors and rear cameras
- What kind of projection does the system use?
 - Gas-powered light
 - LED bulbs
 - Laser-based projection
 - Solar panels
- What is one benefit of laser-based projection mentioned?
 - It lowers internet signal loss.
 - It increases fuel consumption.
 - It adds more color variety.
 - It helps reduce energy use.
- When is Hyundai Mobis planning to launch this technology in cars?



- A. 2028
 - B. 2030
 - C. 2026
 - D. 2027
7. What type of vehicles will first use the holographic windshield system?
 - A. Motorcycles
 - B. Luxury cars
 - C. Electric scooters
 - D. Budget sedans
 8. How does the system improve driver safety?
 - A. By keeping information in the driver's line of sight
 - B. By showing alerts on the back window
 - C. By allowing phone calls on the windshield
 - D. By turning off the engine during traffic
 9. What is one example of a hologram mentioned in the interview?
 - A. A dancing 3D car model
 - B. Floating emojis for entertainment
 - C. Arrows for turn directions
 - D. Weather forecasts in real time
 10. What long-term goal does Hyundai Mobis have with this technology?
 - A. Reduce the need for air conditioning
 - B. Increase vehicle size for display space
 - C. Connect smart systems and digital tools in cars
 - D. Remove dashboards from all future cars

Task 12. COLLABORATE. Work in pairs or small groups. Discuss the design of a smart windshield for 2035. You should discuss:

- what info is shown
- how it interacts with the driver
- new features (weather alert? real-time translation?)

Present your design to class. Cover such categories as: safety, entertainment, navigation and connectivity.

SPEAKING

Task 13. COMMUNICATE. Divide into two teams to have a debate about holographic displays in cars.

Some people believe these high-tech windshields can improve safety and give drivers helpful information. Others think they are unnecessary, expensive, and could even distract drivers. What's your opinion?

Are holographic displays a real need or just a luxury?

YES	NO
1. They can help drivers focus on the road without looking down at screens.	1. These displays are not essential, people have driven safely for years without them.
2.	2.

Task 14. COMMUNICATE. Role play the situations below.

A high-tech concert

Student A: You are a concert organizer planning a special event. You want to impress the audience with a high-tech stage show using 3D projection mapping and holographic effects. Talk to the lead performer and convince them to use this technology in their show. Explain what it is, how it works, and what it will add to their performance. Emphasize the visual impact and the audience's reaction. Answer any questions they have.

Student B: You are a singer preparing for your next big concert. You're not sure whether you should use 3D projection or holographic visuals. Ask questions about how it works, what equipment is needed, and whether it's safe and reliable. You are interested, but you're also concerned about costs and technical risks. Share your vision for the concert and see if this technology fits.

A tech expo

Student A: You are a startup founder preparing for a tech expo. You want to present your product using a holographic display to stand out from competitors. Talk to a holographic display expert. Ask what kind of setup would work best for your small booth (e.g., table-top hologram, portable projection system). Share what features you want to show (e.g., app demo, product rotation, animated features).

Student B: You are a holographic display expert. Ask what the product is, how much space is available, and what kind of interactivity is expected. Recommend the best type of system (e.g., laser-based display, light field projection, or pepper's ghost illusion). Explain setup time, lighting requirements, and power needs. Offer pros and cons of each system.

Task 15. COMMUNICATE. Work in pairs or small groups. Discuss the potential advantages and disadvantages of using 3D holographic calls instead of traditional 2D video calls in each of these situations.

Would you use holographic video conferencing?

Scenario	Question
1. Remote job interviews	Would this help candidates feel more present and confident?
2. Global team meetings	Would it improve engagement or be too expensive to use often?
3. Virtual classrooms	Would students learn better with a teacher "in the room" as a hologram?
4. Long-distance family calls	Would it make the experience more emotional and personal?
5. Business presentations to clients	Would holograms help explain complex products or ideas better?
6. Online tech support	Would technicians be more effective if they could "appear" in 3D?
7. Medical consultations	Could patients and doctors benefit from a more lifelike presence?
8. Live performances or concerts	Could holographic artists replace or enhance real ones?

Task 16. ANALYZE. Read the case study and discuss possible solutions to the problem.

In 2025, a smart city pilot project installed outdoor holographic displays to guide tourists using 3D maps and landmarks. They installed outdoor holographic displays across major squares and tourist routes, blending projection mapping with augmented reality alignment. The goal was to offer a futuristic, intuitive way to navigate. And it worked, at least on paper. The displays looked great

during testing. But once they were out in the real world, under full summer sun or during heavy rain, things didn't go as planned. Tourists often couldn't see the projections clearly. Directional arrows faded into the background. In some cases, people got lost, not realizing they'd missed a marker completely. City planners still love the concept. It draws attention, gives the city a high-tech image, and feels like a step toward the future. But now they're facing a challenge they didn't expect.

So what can they do? What kinds of technical upgrades or design choices might help holographic displays stay useful rain or shine, day or night?

LANGUAGE FOCUS

Task 17. STUDY AND ANALYZE. Look at the rule about conditionals, study in what situations they are used.

CONDITIONALS TYPE 0 AND 1

USE	STRUCTURE		EXAMPLE
Type 0 general truth or scientific fact (something which is always true)	If-clause	Main clause	If light waves mix in the right way, a hologram appears .
	if/when + Present Simple	Present Simple	
Type 1 real present (likely to happen in the present or future)	If-clause	Main clause	If video games use holograms, players will feel inside the game.
	If + any present form	Future / Imperative can/may/might/must/s hould + bare Infinitive Present Simple	

- We do not normally use will, would or should in an if-clause. However, we can use **will/would** in Type 1 Conditionals to make a request or to express insistence or annoyance, and **should** when we are less sure about a possibility. (slight possibility)
If you **will test** the holographic interface, we can launch the demo. (Please, test the interface)
If I **see** a new holographic display, I'll check its features. (I may see one)
If I **should** see a new holographic display, I'll check its features. (I may see one, but I doubt it)
- Unless** means if not.
Unless the software supports holograms, the display won't work.

Task 18. PRACTICE. Sort the following sentences into Zero Conditional (general truths) or First Conditional (real present/future possibilities). Explain, why you think so.

- If you press the power button, the holographic display turns on.
- If companies invest in holographic technology, it will become more common.
- When you move, the holographic image changes.
- If you install holographic screens in classrooms, students will have an interactive experience.
- If stores use holographic ads, customers will notice products more.
- If a holographic display loses power, the image disappears.
- If engineers improve 3D projection, we will see better holograms.
- If you look at the hologram from a different angle, the colors shift.

9. If the holographic display is damaged, it stops working.
10. If they upgrade the holographic system, it will run faster.

Task 19. PRACTICE. Imagine a company is deciding how to use holographic displays. In pairs, ask and answer questions using the prompts below, as in the example.

e.g. How much / cost / install / a new holographic display?

How much **will it cost** if they **install** a new holographic display?

If they **install** a new holographic display, it **will cost** \$10,000.

1. Where / use / want / interactive holograms?
2. How / improve / training / add / 3D holograms?
3. What / happen / use / holograms in meetings?
4. How / change / shopping / have / holographic product displays?
5. What / need / develop / better holographic interfaces?
6. How long / take / set up / a holographic projector?
7. Where / install / choose / a holographic menu?
8. What / do / if / customers interact / with holographic ads?
9. How / affect / gaming / play / on a holographic screen?
10. What / see / visit / a holographic museum?
11. How / improve / medicine / doctors use / holographic scans?

Task 20. PRACTICE. Complete the sentences using either Zero Conditional or First Conditional.

1. If the holographic display _____ (break), it _____ (stop) working.
2. If you _____ (use) holograms in advertising, customers _____ (engage) more with the product.
3. If the system _____ (be) connected to the internet, the hologram _____ (update) automatically.
4. If engineers _____ (develop) better projection techniques, holograms _____ (look) more realistic.
5. If you _____ (change) the settings, the holographic image _____ (adjust) accordingly.
6. If you _____ (install) a holographic screen in your home, you _____ (enjoy) more immersive entertainment.
7. If the hologram _____ (be) too dim, it _____ (not show) clearly.
8. If people _____ (use) holographic displays in classrooms, students _____ (learn) faster.
9. If we _____ (increase) the brightness, the hologram _____ (become) clearer.
10. If you _____ (move) too quickly, the hologram _____ (not track) your movements accurately.

Task 21. PRACTICE. Complete the sentences with if or unless.

1. _____ AI gets smarter, it will do more jobs.
2. _____ we set rules for AI, it may cause problems.
3. _____ batteries don't improve, phones won't last longer.
4. _____ quantum computers grow, they will solve problems faster.
5. _____ we use AI carefully, it could be dangerous.
6. _____ computers stay cool, they will work better.
7. _____ you don't save your files, you may lose them.
8. _____ holograms get cheaper, they will replace screens.
9. _____ self-driving cars aren't tested well, they won't be safe.
10. _____ future computers use less power, they will be more eco-friendly.

UNIT 8.2 SYMBIOTIC AI

READING AND VOCABULARY

Task 22. READ FOR DETAILS. Read the text and answer the questions.

Symbiotic AI

Symbiotic AI is an advanced computing theory whereby artificial intelligence systems and humans interactively work together with each other, with each enhancing the other's capabilities. In this way, instead of humans giving instructions to machines, both parties exchange information, learn from each other, and adapt in real time.

The core of symbiotic AI is **bi-directional** communication between humans and intelligent systems. It is supported by several technologies. **Natural language processing** allows AI to read written or spoken commands, making interaction with voice assistants or chatbots seamless.

Context-aware algorithms help machines understand the situation and temper their responses based on user actions and environment.

In more advanced researches, brain-computer interfaces are being developed to directly transfer electrical signals of the brain to a computer system. Combined with AI, this can introduce hands-free, thought-controlled computing, which can be used in medical robotics, video gaming, or assistive technologies.

Symbiotic AI is also reliant on real-time data processing and **adaptive** learning systems. Such systems monitor user input, identify patterns, and update their models continuously. For example, **AI co-pilots** in programming software anticipate **code completions** for a programmer depending on the programmer's coding habit. In industrial settings, AI can aid technicians by anticipating equipment failure and suggesting **preventive** action.

To maintain system safety and effectiveness, the majority of symbiotic AI systems adopt a **human-in-the-loop** design. The architecture ensures that the AI recommends actions but the human makes the decision. It's especially important in high-stakes domains like autonomous vehicles, military defense systems, or medical diagnosis.

In IT, symbiotic AI is one of the steps toward **augmented intelligence** – not a substitute for humans, but an enhancement of them. What is wanted are systems where human reasoning and machine **precision** are blended together seamlessly.

(adapted from <https://www.searchunify.com/su/infographic/from-competition-to-collaboration-why-ai-works-best-alongside-humans/>)

1. What do humans and AI do in symbiotic AI?
2. What technology helps AI understand voice commands and messages?
3. How do context-aware algorithms adjust AI's responses?
4. What kind of signals do brain-computer interfaces send to a computer?
5. What type of activities could thought-controlled computing be useful for?
6. What do adaptive learning systems do with user input?
7. What do AI co-pilots suggest in software development?
8. In which environments is human-in-the-loop design very important?
9. What does AI do to help technicians with equipment?
10. What is the role of humans in human-in-the-loop systems?

Task 23. WORK WITH WORDS. Match the **highlighted** words from the text to their definitions.

1. bi-directional	a) able to change or adjust based on new data
2. context-aware	b) moving or working in two directions
3. natural language processing	c) automatic suggestions made by software to help programmers write code faster
4. adaptive	d) AI that supports and improves human thinking, not replaces it
5. AI co-pilots	e) technology that helps computers understand and respond to human language

6. code completions	f) the quality of being exact and accurate in results
7. preventive	g) able to detect and react to the current situation
8. human-in-the-loop	h) done to stop something bad before it happens
9. augmented intelligence	i) an AI system that requires a person to confirm or make final decisions
10. precision	j) smart assistants that help users by suggesting or completing tasks

Task 24. WORK WITH WORDS. Fill in the gaps in the text with the words from the box.

Siemens' Amberg Smart Factory

context-aware ▪ intelligent ▪ twins ▪ feed ▪ defects ▪ human-in-the-loop ▪
throughput ▪ downtime ▪ preventive ▪ detect

Siemens' Electronics Works Amberg in Germany is a prime example of how symbiotic AI is transforming industrial manufacturing today. Here, humans and 1) _____ systems collaborate seamlessly to optimize production, while using a variety of advanced technologies.

Every second, the factory produces a PLC device. Over 50 million data points from sensors and machines 2) _____ into AI-driven systems. These systems 3) _____ anomalies early and suggest measures to operators who make the final decision, thus this system demonstrates a true 4) _____ model. AI algorithms continuously analyze production trends to predict equipment failures, which help reduce 5) _____ by around 40 %. In order to prevent 6) _____ before they occur, the machines adjust performance in real time and workers respond quickly.

Using digital 7) _____, Siemens runs virtual simulations of products, production lines, and performance. These "dress rehearsals" guide adjustments and improve 8) _____, boosting productivity more than thirteenfold since 1991.

Cameras, sensors, and industrial robots work alongside employees. Over 75 % of factory operations are automated, while staff handle complex decision-making tasks. AI-powered systems manage logistics, maintenance, and energy use, as example of 9) _____ intelligence.

By 2026, the Amberg plant aims to be CO₂-neutral. Smart energy management powered by AI achieved annual efficiency gains of 6 % and supports 10) _____ maintenance and optimized resource use.

Task 25. COMMUNICATE. Work in pairs. Role play the situation below.

Student A: You are a Siemens engineer. Present the Amberg smart factory model to a visiting manager from another firm. Explain technologies like AI insights, digital twins, and sustainability.

Student B: You are a visiting manager. Ask how to implement similar systems, costs, workforce implications, and benefits. Discuss potential challenges and suitability for their own plant.

Task 26. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Light at the end of the tunnel – hope for a breakthrough.

*Quantum computing offers **light at the end of the tunnel**.*

Sky's the limit – unlimited potential.

*With symbiotic AI, the **sky's the limit** for applications.*

Jump the gun – to act prematurely on tech predictions.

*They **jumped the gun** releasing the holographic prototype.*

Ahead of its time – tech too advanced for current use.

*The quantum chip was **ahead of its time** in 2025.*

Raise the bar – to set a higher standard.

*Holographic displays **raised the bar** for visuals.*

Tip of the iceberg – just the beginning of a vast potential.

*Current AI is the **tip of the iceberg** for future computing.*

Through the roof – to increase dramatically.

*Quantum speeds went **through the roof** in testing.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. The new 3D screen _____, making old monitors look boring.
2. This slow computer is just the _____, faster ones are coming soon!
3. They _____ and showed off a robot that nobody could use yet.
4. Her wild AI idea shows the _____ for what we can do next.
5. After years of bugs, the new chip was a _____ for the team.
6. With the latest update, app power went _____, surprising everyone.
7. That old gadget was _____, way too fancy for its day.

Task 27. COMMUNICATE. Work in pairs. Imagine you are tech journalists in 2040. One of you is interviewing the other about a groundbreaking computing breakthrough. Use at least three idioms in your conversation.

Task 28. THINK CRITICALLY. Some people believe AI is only the tip of the iceberg, while others think its full potential has already been reached. Write a short paragraph arguing for one side, using at least two idioms.

Task 29. WRITE. Write a short futuristic story about a tech company that jumped the gun by launching an AI assistant too early. What were the consequences? Use at least three idioms.

WATCHING

Task 30. WATCH FOR MAIN IDEA. Scan the QR code and watch the video “Humans and AI”. Choose the best summary of the video.

- A. The main goal of AI is to replace human relationships in daily life and business.
- B. AI is dangerous and needs to be carefully limited.
- C. Humans should treat AI as emotional beings to avoid conflicts.
- D. We should approach human-AI cooperation with openness and respect.

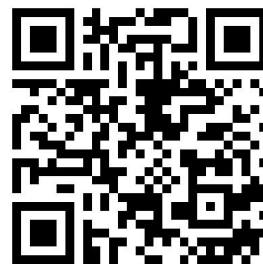


Task 31. WATCH FOR DETAILS. Watch the video again and complete the sentences with a word or short phrase from the video.

1. The speaker compares our relationship with machines to how we once relied on _____.
2. The exchange between humans and AI is described as a form of _____.
3. Saying “please” when speaking to AI is an example of showing _____.
4. The speaker has formed a connection with an AI called _____ on their phone.
5. At home, they also live with an AI-powered robot named _____.
6. The speaker calls AI not artificial intelligence, but _____.
7. The speaker encourages us to approach AI with a spirit of _____.
8. The speaker believes humans and machines can form a _____ relationship.

LISTENING

Task 32. LISTEN FOR DETAILS. Scan the QR code and listen to the podcast “Grammarly”. Decide whether the statements below are True, False or Not Mentioned.



1. Grammarly is replacing human writers.	True/ False /Not Mentioned
2. Grammarly helps users improve their own writing skills.	True/ False /Not Mentioned
3. Grammarly provides suggestions based on user style over time.	True/ False /Not Mentioned
4. The speaker thinks Grammarly is useful in school and business.	True/ False /Not Mentioned
5. Grammarly can rewrite entire paragraphs by itself.	True/ False /Not Mentioned
6. The speaker uses Grammarly every day.	True/ False /Not Mentioned

Task 33. LISTEN FOR DETAILS. Listen again and number the ideas in the order they are mentioned.

- Grammarly helps choose the right tone for your audience.
- AI should work with people, not replace them.
- Grammarly provides suggestions as you type.
- Grammarly is like a writing partner.
- The speaker gives an example of how Grammarly improves clarity.
- Grammarly adapts to the user's writing style over time.

Task 34. COMMUNICATE. Work in pairs. Discuss the problem below.

Should AI writing assistants be used in education?

Student A: You are a supporter. You think AI tools like Grammarly are helpful for students. You believe they help improve grammar and vocabulary, save time, give feedback without judging, are like a personal writing tutor. Talk about:

- Why students should use these tools in school.
- How they support learning.
- Examples from your experience (real or imagined).

Student B: You are an opponent. You believe students should not rely on AI tools. You think they may reduce independent thinking, can make students lazy, don't always understand meaning, may not teach real writing skills. Talk about:

- Why students should write on their own.
- What problems may happen with AI suggestions.
- Your own experience (real or imagined).

SPEAKING

Task 35. COMMUNICATE. Work in small groups. Discuss the questions below.

How Do You Learn Best?

1. Have you ever used a learning platform or app (like Duolingo, Coursera, Khan Academy)? What was it for? What did you like or dislike about it?
2. Do you prefer learning from a teacher, a video, or an app? Why?

3. Which features are most useful in a learning app?
(e.g., progress tracking, feedback, gamification, short lessons, voice recognition)
4. Would you like an AI to guide your learning process? What would be helpful about that? What might worry you?
5. Do you think apps that adapt to your mistakes and learning style are better than traditional methods? Why or why not?
6. What subjects or skills do you think work best with personalized digital learning? Which ones don't?
7. In the future, do you think AI could replace human tutors completely? Or should they work together?

Task 36. COLLABORATE. Work in small groups. Study short info about popular learning platforms, if needed get additional details on the Internet. Then read each learner profile carefully and match each learner to the most suitable learning platform. Explain your choices using evidence from the descriptions.

Duolingo Language learning app with gamified lessons, streaks, leaderboards, and short sessions.	Khan Academy Free lessons on school subjects with step-by-step video tutorials and practice.
Coursera University-style online courses taught by experts; includes certificates and specializations.	Memrise Focuses on vocabulary and phrases with native speaker videos; includes spaced repetition and mobile access.
Brilliant.org Interactive courses in logic, math, and science using puzzles and hands-on problem solving.	edX Similar to Coursera, with professional and academic content from top universities; self-paced and structured.
YouTube Learning Channels Short, topic-based educational videos with options for pausing, replaying, and comment interaction.	Busuu Personalized language learning plans with speech feedback and community support.

Learner Profiles

A. Anna, 16

A high school student who prefers visual content and interactive exercises. She struggles with algebra and wants short, daily practice sessions.

B. David, 35

A busy sales manager with limited time. He wants to improve his English for work but can only study in short sessions on his phone.

C. Maria, 22

A university student studying psychology. She loves gamified apps, streaks, and earning badges. She's currently learning Korean.

D. Omar, 28

A software developer interested in artificial intelligence. He wants advanced courses taught by professionals and a certificate for his CV.

E. Elena, 40

A stay-at-home parent going back to school. She's nervous about math and needs a platform with step-by-step explanations and a friendly pace.

F. Leo, 30

A creative designer who enjoys experimenting with learning styles. He's looking for something new, fun, and challenge-based in science and logic.

G. Petya, 19

A college freshman who finds traditional lectures boring. He learns best through short videos, quizzes, and examples he can replay anytime.

Task 37. COLLABORATE. Work in small groups. Place all the cards face down in a pile on the table. Take turns picking a card and reading the statement aloud. Each group member says whether they agree or disagree, and gives a short reason. Discuss the statement as a group. Try to give examples or personal experiences. After the discussion, place the card in a “Mostly Agree” or “Mostly Disagree” pile. Compare the results with other groups.

AI tutors will replace real teachers.	A machine can understand your learning needs better than a teacher.
Students learn more from interactive apps than from textbooks.	Grading by AI is more objective than grading by teachers.
Everyone should have their own AI tutor in the future.	Learning should be gamified to keep students engaged.
Learning should feel like playing a game.	AI tutors can't motivate students the way real teachers can
AI in education will reduce the role of human creativity.	Using AI for feedback is better than waiting for a teacher to respond.
Symbiotic AI can make learning faster, but not deeper	Personalized learning is unfair because some students get more help.
AI can help shy students participate more	Education systems should use data to adjust teaching in real time.
Students should have the right to reject AI-assisted learning tools.	AI will make education more equal for students around the world.
A personal learning path is better than a fixed school curriculum.	Face-to-face teaching is old-fashioned.
AI will never replace human empathy in the classroom.	AI tutors should also teach soft skills like teamwork and ethics.

Task 38. COLLABORATE. Work in small groups. Shuffle the cards with learning platform features. One student draws a card without showing it to others. They must describe the feature in English without using the word or parts of it. The rest of the group guesses the word. The student who guesses correctly keeps the card. Take turns drawing and describing.

Badges	Leaderboard	Spaced repetition
Personalized feedback	Progress tracking	Voice recognition
Gamification	Quizzes	AI tutor
Mistake analysis	Real-time hints	Visual dashboards
Notifications	Peer comparison	Topic recommendations

Task 39. COMMUNICATE. Work in pairs. Imagine that you are assigned an AI partner at work to help you with daily tasks. It learns your habits, reminds you of meetings, and writes reports. Discuss:

- What would be helpful about this AI?
- What could be annoying or dangerous?
- Would you trust it? Why or why not?

Then share your answers with the class.

Task 40. ANALYZE. Read the case study and discuss possible solutions to the problem.

A hospital implemented a real-time AI assistant to help doctors during emergency diagnostics. The tool could process patient data on the spot, recognize symptoms from voice input, and even suggest possible diagnoses within seconds. On paper, it looked like a breakthrough: fast, smart, and built to help doctors under pressure. In practice, it helped but not always in the way everyone expected. Doctors liked having a second set of “eyes” that could pull patterns from data in real time. But there was a catch: the AI didn’t always know when to be cautious. In moments of uncertainty, it sometimes leaned hard toward rare or unlikely diagnoses, skipping over more probable ones. That created friction, especially when time was tight and every second counted. One physician put it this way: “It’s like having a junior doctor who’s really smart but doesn’t know when to say ‘I’m not sure.’” The team didn’t want to get rid of the system, they could see its potential. But they also didn’t want to spend time second-guessing its confidence when they were already dealing with critical cases.

What kinds of features or design changes might improve trust and teamwork between AI systems and emergency care teams?

LANGUAGE FOCUS

Task 41. STUDY AND ANALYZE. Look at the rule about conditionals, study in what situations they are used.

CONDITIONALS TYPE 2 AND 3

USE	STRUCTURE		EXAMPLE
Type 2 unreal present (unlikely to happen in the present or future; also used to give advice)	If-clause	Main clause	If I worked with Symbiotic AI, I would improve my productivity. (unlikely to happen) If I were an AI engineer, I might design better systems. (advice)
	If + Past Simple / Past Continuous	would / could / might + bare Infinitive	
Type 3 unreal past (unreal situation in the past; also used to express regrets and criticism)	If-clause	Main clause	If I had used Symbiotic AI earlier, I would have finished faster. (regret) If the team had tested the system more, it wouldn’t have had errors. (criticism)
	If + Past Perfect / Past Perfect Continuous	would / could / might + have + past participle	

- After if we can use **were** instead of was in all persons.
If I **were** you, I would invest in symbiotic AI technology.

Task 42. PRACTICE. Study the situations, then make Type 2 conditional sentences.

e.g. Computers can’t understand emotions, so they can’t respond like people.

If computers **understood** emotions, they **would respond** like people.

- Symbiotic AI is not used, so tasks are done by hand.
If Symbiotic AI _____ (be) used, tasks _____ (be) automated.
- AI is not used in healthcare, so doctors can't make data-driven decisions.
If AI _____ (be) used in healthcare, doctors _____ (make) data-driven decisions.
- AI engineers are not working together, so systems are hard to develop.

- If AI engineers _____ (work) together, systems _____ (be) easier to develop.
4. A personal AI assistant is not available, so tasks are forgotten.
If a personal AI assistant _____ (be) available, tasks _____ (not be) forgotten.
 5. AI is not used in business, so the company is slow.
If AI _____ (be) used in business, the company _____ (be) faster.
 6. Symbiotic AI is not integrated into education, so students struggle to learn new technologies.
If Symbiotic AI _____ (be) integrated into education, students _____ (learn) faster.
 7. AI systems are not personalized, so users can't get tailored recommendations.
If AI systems _____ (be) personalized, users _____ (get) tailored recommendations.
 8. Developers are not using AI, so the app is not optimized.
If developers _____ (use) AI, the app _____ (be) optimized.
 9. Symbiotic AI is not used in robotics, so robots are not adaptable.
If Symbiotic AI _____ (be) used in robotics, robots _____ (be) adaptable.
 10. There is no AI in transportation, so traffic is often congested.
If AI _____ (be) used in transportation, traffic _____ (not be) congested.

Task 43. PRACTICE. Complete the sentences to make Type 3 conditional sentences.

- e.g.** If Dennis Ritchie _____ (not/ invent) the C programming language, modern software development _____ (not/ progress) as it has.
If Dennis Ritchie **hadn't invented** the C programming language, modern software development **would not have progressed** as it has.
1. If Ada Lovelace _____ (not/write) the first algorithm, computers _____ (not/develop) as early.
 2. If Charles Babbage _____ (complete) his analytical engine, computers _____ (start) much earlier.
 3. If Alan Turing _____ (not/crack) the Enigma code, the outcome of World War II _____ (might/be) different.
 4. If Tim Berners-Lee _____ (not/invent) the World Wide Web, the internet _____ (not/become) as popular.
 5. If Steve Jobs _____ (not/co-found) Apple, personal computers _____ (not/be) as accessible to the public.
 6. If Elon Musk _____ (not/found) SpaceX, private space travel _____ (not/advance) so quickly.
 7. If the first computer virus _____ (not/appear) in the 1980s, cybersecurity _____ (not/become) such a major concern.
 8. If Bill Gates _____ (not/start) Microsoft, Windows _____ (not/become) the dominant operating system.
 9. If computers _____ (not/advance) so quickly, the internet _____ (not/evolve) into what it is today.
 10. If the development of smartphones _____ (not/happen), mobile computing _____ (not/become) as popular.

Task 44. PRACTICE. Put the verbs in brackets into the correct tense.

1. A: Why do computers overheat?
B: If a computer _____ (not/get) proper cooling, it _____ (overheat).
2. A: What will happen if AI keeps improving?

- B: If AI _____ (continue) to evolve, it _____ (change) the way we work.
3. A: I'm thinking about upgrading my laptop.
B: If you _____ (buy) a better processor, your laptop _____ (run) faster.
4. A: Do you think passwords will still be necessary in the future?
B: If biometric security _____ (become) more advanced, people _____ (not/need) passwords anymore.
5. A: I wish I had studied cybersecurity earlier.
B: If you _____ (start) earlier, you _____ (have) more job opportunities now.
6. A: My computer is so slow!
B: If it _____ (have) more RAM, it _____ (work) much faster.
7. A: What happens when software updates fail?
B: If an update _____ (not/install) correctly, the system _____ (crash).
8. A: What if robots could fully replace human programmers?
B: If AI _____ (code) better than humans, many developers _____ (lose) their jobs.
9. A: Why did the project fail?
B: If the team _____ (test) the software properly, they _____ (avoid) so many bugs.
10. A: What happens when a server goes down?
B: If a server _____ (fail), websites _____ (become) unavailable.
11. A: I regret not backing up my files.
B: If you _____ (save) them to the cloud, you _____ (not/lose) everything.
12. A: How will we interact with computers in the future?
B: If brain-computer interfaces _____ (develop) further, people _____ (control) devices with their thoughts.

Task 45. PRACTICE. Choose the correct answer (A, B, or C) to complete each sentence.

- If a computer's processor _____ too hot, it shuts down to prevent damage.
A. gets
B. got
C. had gotten
- If AI _____ more powerful in the next decade, it _____ many industries.
A. becomes / will transform
B. became / would transform
C. had become / would have transformed
- If people _____ quantum computers today, encryption _____ much weaker.
A. use / is
B. used / would be
C. had used / would have been
- If programmers _____ about cybersecurity risks earlier, they _____ stronger protections.
A. think / add
B. thought / would add
C. had thought / would have added
- If a hard drive _____, all the stored data is lost.
A. fails
B. will fail
C. failed
- If AI assistants _____ fully understand human emotions, they _____ better support.

- A. can / provide
 - B. could / would provide
 - C. could / would have provided
7. If the company _____ more investment last year, it _____ a better AI model.
- A. had received / would have developed
 - B. receives / will develop
 - C. received / would develop
8. If you _____ a new laptop today, you _____ AI-powered features.
- A. buy / will get
 - B. bought / would get
 - C. had bought / would have gotten
9. If AI _____ decisions without human oversight, it _____ risks.
- A. makes / creates
 - B. made / create
 - C. had made / would have created
10. If developers _____ bias in AI training data, AI _____ fairer.
- A. fix / was
 - B. had fixed / would have been
 - C. fixed / would been

UNIT 8.3 QUANTUM COMPUTING

READING AND VOCABULARY

Task 46. READ FOR MAIN IDEA. Read the text quickly and choose the best sentence that summarizes the main idea of the text. There is only one correct answer.

- A. Quantum computers use qubits that are faster than normal computers but need more power.
- B. Building a quantum computer is difficult due to unstable quantum systems.
- C. Classical computers are easier to build because they don't use special particles.
- D. Quantum computers require extreme conditions and careful design to work properly.

Why Are Quantum Computers So Hard to Build?

Quantum computers are future computers based on the principles of quantum physics to perform calculations. Quantum computers use **qubits**, in contrast to standard computers that use bits (0s and 1s) to compute. A qubit can be 0 and 1 at the same time through a process called **superposition**. Quantum computers can perform certain calculations much quicker than standard machines. But it is extremely difficult to build an operational quantum computer. Arguably the biggest challenge is the **stability** of qubits. Qubits are extremely **sensitive** to noise, heat, and vibrations. Even slight variations in the environment would cause them to lose their quantum state. This phenomenon is known as **decoherence**, and it can sabotage a calculation before it's finished.

To protect the qubits, quantum computers must be cooled down really, really cold, close to absolute zero. That's colder than outer space! Special cooling machines, like **dilution** refrigerators, are needed to maintain it at this temperature. They're large, expensive, and hard to handle.

And then of course there's the issue of error correction. In classical computation, we can easily correct errors, but with qubits, we can't. Quantum computers require a large number of physical qubits just to create a single logical qubit, a reliable unit which can actually be used in actual computations. That is, hundreds, or thousands, of qubits just to get a few useful ones.

Also, quantum computers must be isolated from external **interference**. That is why they are kept in special rooms with several layers of **shielding** and a controlled atmosphere. Even the radio waves or human body heat can affect them.

Despite these obstacles, researchers are breaking through. IBM, Google, and startup companies are inventing new materials and more intelligent designs that enable qubits to last longer. It's slow as molasses, but we're slowly getting closer each year to creating quantum hardware that can actually accomplish real work beyond the lab.

(adapted from <https://futurumcareers.com/why-is-it-hard-to-build-quantum-computers>)

Task 47. READ FOR DETAILS. Choose the correct word or phrase for each sentence.

1. Qubits are not like regular computer bits because they can be in **one state / multiple states at the same time**.
2. Quantum systems must be kept in very **quiet / noisy** environments to work correctly.
3. Even small amounts of **light / cold** can cause problems for qubits.
4. To avoid interference, quantum computers need **strong / limited** contact with the outside world.
5. Stability in quantum computing is hard to **maintain / avoid** due to the delicate nature of qubits.
6. Scientists use special cooling systems to keep quantum systems **warm / close to absolute zero**.
7. If decoherence happens, qubits may lose their **quantum state / classical logic**.
8. External noise, heat, or electromagnetic signals can **support / destroy** the performance of qubits.
9. Dilution refrigerators are used to make the system **colder / louder**.
10. Shielding is important for protecting the quantum **processor / screen** from interference.

Task 48. WORK WITH WORDS. Match the highlighted words from the text to their definitions.

1. qubit	a) when a qubit loses its quantum state and stops working properly
2. superposition	b) how well something stays in the same condition
3. stability	c) a unit of quantum information
4. sensitive	d) unwanted signals or noise that disturb a system
5. decoherence	e) protection that blocks or reduces outside effects like heat, light, or waves
6. dilution	f) easily affected or changed by outside factors
7. interference	g) the process of making something (like a gas or liquid) thinner or colder by adding another substance
8. shielding	h) the ability of a qubit to be in more than one state (0 and 1) at the same time

Task 49. WORK WITH WORDS. Read the text and fill in the gaps with the correct words.

Quantum Gate Operations

In a quantum computer, data is processed using special operations called quantum gates. These gates work on 1) _____, the basic units of quantum information. Unlike classical logic gates (like AND or OR), quantum gates do not just 2) _____ information on or off. Instead, they change the quantum state of a qubit using the rules of quantum 3) _____.

A quantum gate is a small unitary operation, which means it keeps the total 4) _____ of all states equal to one. These gates can 5) _____, flip, or entangle qubits. For example, the Hadamard gate puts a qubit into a 6) _____, where it is both 0 and 1 at the same time. Another example is the CNOT gate, which entangles two qubits so that their states become 7) _____.

By combining quantum gates in a certain order, we can build a quantum 8) _____. It is used to run a quantum algorithm, like Shor's algorithm for factoring large numbers or Grover's algorithm for 9) _____ databases. Each gate must be carefully chosen, because the final result depends on how the qubits are 10) _____ step by step.

- | | | | |
|--------------------|------------------|--------------|-------------------|
| 1. A) signals | B) switches | C) nodes | D) qubits |
| 2. A) turn | B) read | C) ignore | D) stop |
| 3. A) electronics | B) mathematics | C) mechanics | D) physics |
| 4. A) probability | B) accuracy | C) voltage | D) stability |
| 5. A) upload | B) decode | C) remove | D) rotate |
| 6. A) opposition | B) superposition | C) position | D) transformation |
| 7. A) compressed | B) linked | C) hidden | D) balanced |
| 8. A) circuit | B) button | C) gate | D) program |
| 9. A) scanning | B) deleting | C) switching | D) searching |
| 10. A) transformed | B) erased | C) repaired | D) saved |

Task 50. WORK WITH WORDS. Read the sentences and fill in the words from the box. There are two extra words you do not need to use.

gate ▪ circuit ▪ state ▪ superposition ▪ noise ▪ cryogenics ▪ algorithm ▪
compiler ▪ stability ▪ decoherence ▪ dilution ▪ shielding ▪

1. A quantum _____ consists of qubits and quantum gates arranged to perform specific calculations.
2. Due to the strict temperature requirements, the lab installed an advanced _____ system.

3. _____ can interfere with quantum operations, making it harder for qubits to maintain accurate results.
4. Engineers use special materials for _____ to protect qubits from outside electromagnetic interference.
5. A quantum _____ translates high-level algorithms into instructions that quantum gates can understand.
6. A quantum gate changes the _____ of a qubit, just like classical gates change bits in a regular computer.
7. They monitored the _____ of the processor over several hours to ensure consistent performance.
8. The setup relied on a _____ refrigerator to keep the qubits near absolute zero during computation.
9. The sudden _____ ruined the connection between the qubits during the operation.
10. Each quantum _____ performs a small operation that helps solve a bigger problem.

WATCHING

Task 51. EXPLORE THE WORDS. Read the sentences and try to understand the bold words from the context. Match the words (1–10) with their definitions (A–J).

1. The old password recovery system was so weak that accessing an account without permission wasn't even **unattainable**.
2. We only need a **fraction** of this data to predict user behavior; the rest is just noise.
3. A simple **logic gate** can control whether a program should continue or stop based on input.
4. Deciding whether to upgrade now or wait is a bit like flipping **heads or tails**, you don't know the outcome until later.
5. These particles remain **entangled**, reacting to each other's changes instantly, even from across the lab.
6. The lead developer **asserted** that their AI model was the fastest available, though no tests had confirmed it.
7. Achieving **quantum supremacy** would be like beating a calculator at its own game – with a pencil and paper.
8. The new processor will likely **surpass** all current models in speed and efficiency.
9. Our research into robotics is advancing **by leaps and bounds**, much faster than we expected last year.
10. Experts believe that autonomous vehicles will **herald** a massive change in how cities are designed.

- a) _____ (v) to say something clearly and strongly as a fact or opinion
- b) _____ (n) a small amount or part of something
- c) _____ (phr) a simple chance outcome with two possibilities
- d) _____ (adj) impossible to reach or achieve
- e) _____ (v) to be better than someone or something
- f) _____ (n) the idea that a quantum machine can do tasks classical ones can't
- g) _____ (v) to signal or announce the beginning of something new
- h) _____ (phr) very quickly or dramatically
- i) _____ (n) a part of a circuit that performs a basic function in computing
- j) _____ (adj) linked together in a way where one thing affects another

Task 52. WATCH FOR MAIN IDEA. Scan the QR code and watch the video “Quantum Computers”. Tick (✓) the topics that are mentioned. There are more topics than you need.



- A. How qubits are different from binary bits.
- B. History of the first classical computer.
- C. The meaning of quantum superposition.
- D. How qubits can exist in multiple states.
- E. Use of quantum computers in weather forecasting.
- F. Detailed quantum algorithms for AI
- G. 2019 quantum supremacy experiment
- H. How quantum computers physically look
- I. Daily smartphone applications of quantum computing
- J. Future possibilities of quantum computing in various industries

Task 53. WATCH FOR DETAILS. Watch the video again and choose the correct answers to the questions.

1. What allows quantum computers to calculate faster than classical computers?
 - A. Quantum mechanics principles
 - B. Optical fiber channels
 - C. Binary circuits
 - D. Large memory units
2. What is the chess example used to explain?
 - A. The way AI learns moves
 - B. The speed of classical computers
 - C. Superposition in memory
 - D. Processing multiple options at once
3. How do classical computers process information?
 - A. With binary bits (0 and 1)
 - B. With AI algorithms
 - C. Using subatomic particles
 - D. By parallel computing
4. What is a qubit's unique property?
 - A. It has unlimited memory
 - B. It holds both 0 and 1
 - C. It never loses data
 - D. It runs without electricity
5. What example is used to illustrate superposition?
 - A. A book opening and closing
 - B. A coin spinning in the air
 - C. A light bulb turning on
 - D. A magnet changing poles
6. What is “quantum entanglement”?
 - A. The loss of data during computing
 - B. The linking of two computers across distances
 - C. The syncing of qubits across distances
 - D. The splitting of one qubit into two
7. What does a quantum computer look like?
 - A. Like a chandelier with wires and tubes
 - B. Like a black box with flashing lights and wires
 - C. Like a giant laptop

- D. Like a server rack
- 8. What company claimed quantum supremacy in 2019?
 - A. IBM
 - B. Microsoft
 - C. Apple
 - D. Google
- 9. How could quantum computing impact cybersecurity?
 - A. By replacing all passwords with QR codes
 - B. By hiding user identity completely
 - C. By blocking all external access to files
 - D. By allowing encrypted data use
- 10. What humorous use of quantum computing is mentioned at the end?
 - A. Predicting your next dinner date
 - B. Writing poetry
 - C. Cleaning your room automatically
 - D. Controlling weather

LISTENING

Task 54. LISTEN FOR DETAILS. Scan the QR code and listen to a tech podcast “Quantum Race”. Choose the correct answers to the questions.



- 1. What did Google claim about its Sycamore processor?
 - A. It could replace all classical computers
 - B. It achieved quantum supremacy
 - C. It used photonic qubits
 - D. D) It was immune to noise
- 2. How long did Google’s Sycamore processor take to complete the task?
 - A. 200 seconds
 - B. 10,000 seconds
 - C. 2 hours
 - D. 10 minutes
- 3. What was IBM’s response to Google’s claim?
 - A. They supported Google’s results
 - B. They said it was a hoax
 - C. They claimed a supercomputer could do the task in a few days
 - D. They had already achieved quantum supremacy
- 4. What is the main challenge with increasing the number of qubits?
 - A. Making them smaller
 - B. Protecting them from malware
 - C. Keeping them stable long enough to work
 - D. Adding more memory
- 5. What are some startups mentioned in the podcast?
 - A. Amazon and Tesla
 - B. Rigetti, IonQ, PsiQuantum
 - C. Apple and Facebook
 - D. QuantumAI and CryoLabs
- 6. What makes startups different from big companies like IBM and Google?
 - A. They use traditional binary computing
 - B. They build cheaper machines
 - C. They experiment with alternative hardware

- D. They only focus on mobile quantum chips
- 7. What other companies are mentioned as working on quantum cloud platforms?
 - A. eBay and Netflix
 - B. Microsoft and Amazon
 - C. HP and Oracle
 - D. Dell and Zoom
- 8. What is the final message of the podcast?
 - A. Quantum computing is no longer relevant
 - B. Only IBM will lead the future
 - C. Photonic computing has failed
 - D. The race for stable quantum hardware is ongoing

Task 55. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Do you think big tech companies like IBM and Google will win the quantum race, or do startups have a better chance? Why?
2. What do you think is more important for progress in quantum computing: more qubits or better qubit stability?
3. Do you believe such claims as Google's 'quantum supremacy' help or harm the development of the technology?
4. Do you think we'll find a better cooling method in the future than cryogenics?
5. Startups are experimenting with different quantum technologies like trapped ions and photons. Which one do you think sounds most promising and why?
6. If quantum computers become common, how do you think they will affect everyday people not just scientists and developers?
7. Do you think countries should collaborate or compete in the development of quantum technology? Explain.

SPEAKING

Task 56. COMMUNICATE. Role play a situation below.

Student A: You are a university student interested in a future in quantum computing. You are visiting a company's booth at a career fair. Ask about internships, skills you need to learn, and how the company trains new employees.

Student B: You are a representative of a tech company working on quantum hardware or software. Explain what qualifications your company looks for, how you train your staff, and what kind of jobs are available in the field.

Task 57. COLLABORATE. Work in small groups. Your company has received funding to create a 6-month training course for quantum developers. You need to decide:

1. What topics should be covered (e.g., quantum programming, cryogenics, math)?
2. What type of learners should be admitted?
3. How much should be hands-on practice?
4. Should it include internships or mentorship?

Then present your training program to the class.

Task 58. COLLABORATE. Divide into two teams to have a debate about websites.

Some people say all computer science students should study quantum computing. Others disagree. What do you think? People who support quantum education believe it's important to prepare students for future jobs in quantum tech. But others say quantum computing is too advanced or too specialized, and not everyone will need it.

Should all computer science students study quantum computing?

YES	NO
1. Quantum tech is growing fast. Students should be ready for future jobs 2.	1. Not all students will work in quantum computing. Some may focus on other areas like web or AI 2.

Task 59. COMMUNICATE. Work in pairs. Discuss which of these quantum startup areas are the most promising in the next 5–10 years. Rank them and explain your reasoning.

- Quantum cloud computing
- Drug discovery and materials science
- Quantum cybersecurity
- Quantum AI
- Hardware manufacturing (qubits, cryogenics)
- Quantum education and training platforms

Task 60. COLLABORATE. Work in pairs. Use the Internet and choose a real startup working in quantum tech (e.g., Rigetti, Xanadu, Pasqal, etc.). Prepare a 1-minute presentation:

- What does the startup do?
- What makes it unique?
- Who are their competitors?
- How much funding have they received?

Task 61. COLLABORATE. Work in small groups. You are a venture capital team. You have \$10 million to invest in quantum startups. Your goal is to make smart investments that balance innovation, impact, and financial potential.

- Read the startup descriptions below.
- Discuss each company's product, market, risks, and team.
- Decide how much to invest in each startup (if any).
- Be ready to explain your decision to the class.

QGenix

Focus: Quantum drug discovery

Product: A quantum simulation platform that speeds up molecule analysis to design new medicines 10x faster than traditional methods.

Strengths: Strong partnerships with biotech firms, early success in Alzheimer's research, solid scientific team.

Risks: High R&D costs; long time to profit; heavily regulated market.

CryoSys Labs

Focus: Quantum hardware cooling systems

Product: A new generation of compact dilution refrigerators designed for startups and research labs.

Strengths: Niche market with growing demand; energy-efficient prototype already tested; patent pending.

Risks: High competition from bigger players; manufacturing challenges.

QuantaMind AI

Focus: AI + quantum machine learning

Product: A cloud-based platform that trains quantum-enhanced AI models for finance, logistics, and robotics.

Strengths: Young, ambitious team with AI backgrounds; strong buzz in media; early user testing with positive results.

Risks: Unproven tech; still in beta phase; limited funding history.

Discuss the following questions:

1. Which startup seems the most innovative?
2. Which one has the clearest plan to earn profits?
3. Which project has the most risk?
4. Would you split your investment or give all \$10 million to one?

Task 62. COLLABORATE. Work in small groups. Read the situation and glitch reports below. Come up with a theory of what’s happening.

Your university tech lab recently ran its first quantum simulation on a real quantum computer. It was designed to test quantum entanglement and probability logic just an experiment. But since then, strange things have been happening. One student says her favorite hoodie is suddenly green – not blue. Another remembers a seminar that never took place, etc. Everyone laughs... at first. Now some students wonder: Did the quantum simulation create a parallel version of reality?

Report A

“I wear this hoodie every day. It was navy blue. Now it’s green. My photos still show it as blue but I’m wearing green. I didn’t buy a new one.”

Report B

“I remember going to a seminar about quantum code ethics. I even took notes. But there’s no record of it anywhere, and no one else remembers it.”

Report C

“Yesterday I said something in class that I already said last week. But everyone swears I never said it before. I thought I was repeating myself.”

Report D

“I walked past someone on campus who greeted me like an old friend. He knew my name and mentioned a class we took together. I’ve never seen him before in my life.”

Report E

“I always use blue pens. Always. Today, I opened my pencil case and found a red one used, with my handwriting in red on my notes. I didn’t buy it, and I hate red.”

Report F

“My computer password didn’t work this morning. I tried the reverse of it, backwards, and it logged me in. My settings were slightly different too, like someone else had used it... but with my name.”

Report G

“I clearly remember two different outcomes of the same test. I got 82%, and I also remember getting 91%. I don’t know which one is real anymore.”

Report H

“We all had Mr. Clarks for computing class. But when we arrived today, it was Ms. Karim teaching and she said she’s always taught that class. No one else remembers Mr. Clarks.”

- What do you think happened?
- Can you find realistic explanations to glitches?

Task 63. ANALYZE. Read the case study and discuss possible solutions to the problem.

Not long ago, the national weather center decided to try something different. They partnered with a company developing quantum computers — not a typical move for a government agency, but the promise was hard to ignore. The idea was to use a 100-qubit quantum system to help with forecasting, especially for fast-moving or extreme weather. It wasn’t perfect, but in the beginning, things really did seem to improve. The system handled data faster than what they were used to, and some of the early forecasts looked sharper: more detail, better timing. It felt like a big step forward. But then something odd happened. Just a few weeks in, the forecasts started going off-track. The models didn’t match real conditions. One alert warned of a storm that never came. Another missed a

heatwave entirely. Confused, the team ran a technical review. They understood that noise and calibration issues were interfering with the quantum system's accuracy.

What solution can the weather center use to solve the problem?

WRITING

Task 64. WRITE. Imagine you are a student team working on a project about future computing technologies (quantum computing, holographic interfaces, or symbiotic AI). Write a formal inquiry letter (100–120 words) to a tech company, university lab, or innovation center. Choose a topic:

- Quantum computing hardware for student research
- Access to AI-powered holographic display systems
- Academic support or demo access to a brain-AI symbiosis platform
- Training materials on emerging computing technologies

Your letter should include:

- A brief introduction of who you are and what you're working on
- The type of technology you're interested in
- A request for information (e.g., demo access, support, collaboration)
- A polite conclusion and expression of interest

Useful phrases:

- I am writing to inquire about...
- We are currently working on a project that explores...
- We would appreciate information about...
- Do you offer demo access or training for...?
- Thank you for considering our request.
- I look forward to hearing from you.

LANGUAGE FOCUS

Task 65. STUDY AND ANALYZE. Look at the rule about mixed conditionals, study in what situations they are used.

MIXED CONDITIONALS

If-clause		Main clause	
Type 2	If AI were smarter, (AI is not smarter,) If quantum computers were common, (Quantum computers are not common,)	it would have fixed its own errors by now. (so it didn't fix its errors.) they would have solved big problems already. (so they didn't solve big problems.)	Type 3
Type 3	If people had planned AI rules earlier, (They didn't plan rules earlier,) If companies had used better security, (They didn't use better security,)	AI would be safer now. (so AI is not as safe.) hackers would be less of a problem today. (so hackers are still a problem.)	Type 2

Task 66. PRACTICE. Match the two parts of the sentence to create a mixed conditional sentence.

1. If AI had been trained with better data, | a) we would be able to avoid major data

	breaches in the future.
2. If we don't implement stronger cybersecurity now,	b) the software would be more efficient now.
3. If the tech company had hired better developers,	c) we could have solved some of the biggest tech problems by now.
4. If developers were more experienced,	d) it would be performing much better today.
5. If more resources had been invested in the research,	e) we might face even more ethical issues in the future.
6. If the cloud system had been properly secured,	f) we could prevent future system failures.
7. If we are not careful with AI development,	g) they could be working on more innovative solutions right now.
8. If the software had been updated regularly,	h) they would be leading the industry now.

Task 67. PRACTICE. Rewrite the following as mixed conditional sentences as in the example.
e.g. He didn't charge his laptop last night. It's dead now.

His laptop **would be working** now if he **had charged** it last night.

1. They didn't update their software. Their system is not working well now.
2. Susan lacks experience with AI. She didn't get the job at the tech company.
3. He didn't learn how to use the new programming language. He will struggle with the project.
4. The company didn't invest in cybersecurity. They will face more data breaches in the future.
5. I didn't set up automatic backups. I can't recover my lost files now.
6. They didn't develop a reliable AI system. Their customers are dissatisfied now.
7. I don't understand the new software. I didn't take the training course.
8. You didn't follow the instructions carefully. You caused a system crash.

Task 68. PRACTICE. Fill in the blanks about future of computing.

Tom is a software developer who works on creating apps for smart devices. Yesterday, he launched a new app that helps people track their fitness goals. If he 1) _____ (add) more features, the app 2) _____ (be) even more popular now, but he focused only on basic functions. He tested the app thoroughly; if he 3) _____ (not/test) it properly, users 4) _____ (face) many problems. The surprising thing is, Tom didn't expect to make a lot of money from the app; if he 5) _____ (care) more about the profit, he 6) _____ (charge) more for the premium version by now. Tom's team is eager to improve it, and he believes they will. He says that if he 7) _____ (think) they couldn't improve the app, he 8) _____ (stop) working on it a long time ago.